Formative Assessment 1 Marking Report

Software Development Fundamentals

1 Assessment Summary

Student Name	Charlean du Plessis
Student Number	
Assessment Title	Software Development Fundamentals - Formative Assessment 1
Total Marks Obtained	141
Total Marks Possible	160
Percentage	(141 / 160) x 100 = 88.13%
Result	PASS

Submission completeness: 92-page assessment PDF, two Java files, SQL, README, input files, generated CSV, Testing_Data notes and completed Excel test table reviewed. No missing assessment page detected.

Marks by Question

Question	Marks Awarded	Marks Possible
Q1	11	12
Q2	15	16
Q3	15	18
Q4	29	32
Q5	15	18
Q6	20	24
Q7	22	22
Q8	14	18
TOTAL	**141**	**160**

Marking basis: No official memorandum supplied. Expected criteria are derived from the question paper. Original subquestion allocations are retained; partial credit reflects demonstrated technique, completeness and functional evidence. Pass threshold: 50% as printed on the assessment.

2 Question by Question Marking Table

Source files and evidence were considered together. Valid alternative implementations were accepted. Hard-coded results were not credited where the brief requires processing live input or records. Only concepts required by the supplied question paper were assessed.

Question	Student Answer Summary	Expected Answer / Marking Criteria	Marks Awarded	Feedback
1(a)	Named IDE/project evidence.	Project screenshot and logical named SkillsWorkshopManagementSystem project structure.	2/2	Requirement met in the submitted code and/or relevant evidence. No deduction.
1(b)	Java startup file supplied or shown.	WorkshopApplication.java contains the application startup code.	2/2	Requirement met in the submitted code and/or relevant evidence. No deduction.
1(c)	Valid entry-point code.	public static void main(String[] args) starts the application.	2/2	Requirement met in the submitted code and/or relevant evidence. No deduction.
1(d)	Complete named/versioned banner with a literal date.	Application, developer, version, current date and welcome.	1/2	1/2: the source prints 30 September 2026 instead of obtaining the current date dynamically.
1(e)	Required console menu displayed.	All nine required menu functions are displayed.	2/2	Requirement met in the submitted code and/or relevant evidence. No deduction.
1(f)	Source/output evidence of startup and selection.	Compile, start, show heading/menu and accept a choice.	2/2	Requirement met in the submitted code and/or relevant evidence. No deduction.
Q1 total	Sum of assessed subquestions.	Question weighting: 12 marks.	**11/12**	2 + 2 + 2 + 1 + 2 + 2 = 11.
2(a)	Required typed fields and availability calculation.	Workshop/participant fields, meaningful constant; spaces = capacity - registrations.	4/4	Requirement met in the submitted code and/or relevant evidence. No deduction.
2(b)	Reusable string helpers and input/output evidence.	Trim, format names, lowercase/validate email, partial name search and case-insensitive comparison; demonstrate I/O.	4/4	Requirement met in the submitted code and/or relevant evidence. No deduction.
2(c)	Fixed title array with selection and searching.	At least five titles; loop display, user selection, bounds validation and exact/partial array search.	4/4	Requirement met in the submitted code and/or relevant evidence. No deduction.
2(d)	Enum selection/storage works; filtering uses the fixed standard-title/category arrays.	Five required values; user selection, storage/display and category filtering.	3/4	3/4: the filter does not search the newly captured workshop records. Apply it to the maintained records and their selected categories.
Q2 total	Sum of assessed subquestions.	Question weighting: 16 marks.	**15/16**	4 + 4 + 4 + 3 = 15.
3(a)	Menu repetition with an exit condition.	Continue after an operation until the user chooses Exit.	3/3	Requirement met in the submitted code and/or relevant evidence. No deduction.

Question	Student Answer Summary	Expected Answer / Marking Criteria	Marks Awarded	Feedback
3(b)	Menu actions dispatched by switch.	Appropriate cases, actual operations and helpful default handling.	3/3	Requirement met in the submitted code and/or relevant evidence. No deduction.
3(c)	Selection structures apply practical rules.	Appropriate if/else selection for validations and registration conditions.	3/3	Requirement met in the submitted code and/or relevant evidence. No deduction.
3(d)	Loops process records and perform arithmetic; the summary asks for five new sets of values.	Display/search records, retry input and calculate registration/space/income totals.	2/3	2/3: summary totals should be calculated from stored workshops/registrations rather than separately re-entered capacity, booking and fee values.
3(e)	ID, duplicate, active and full-workshop checks supplied.	Valid fields, existing IDs, active/future/not-full workshop and no duplicate booking.	2/3	2/3: registration does not recheck the stored workshop date. A date accepted at capture can later be past; the SQL route also bypasses the booking rules.
3(f)	Local delete, cancel, clear and unsaved-exit prompts are shown.	Confirm deletion, cancellation, clearing and exit with unsaved information.	2/3	2/3: SQL cancellation has no confirmation, and local clearing resets counters while retaining collection/set contents. Confirm every destructive route and perform the complete operation.
Q3 total	Sum of assessed subquestions.	Question weighting: 18 marks.	**15/18**	3 + 3 + 3 + 2 + 2 + 2 = 15.
4(a)	Operational generic lists, maps and sets.	Use ArrayList, HashMap and another suitable generic collection in the application.	4/4	Requirement met in the submitted code and/or relevant evidence. No deduction.
4(b)	Nine workshop fields are stored as records and in arrays.	Maintain nine fields: ID, title, category, facilitator, date, fee, capacity, registrations and active flag.	3/4	3/4: the workshop string held in the collection/map is a capture-time snapshot; its registration count is not maintained after bookings.
4(c)	Participant records represented in collections.	Maintain seven fields: ID, first name, surname, email, telephone, type and active flag.	4/4	Requirement met in the submitted code and/or relevant evidence. No deduction.
4(d)	Efficient keyed record lookup.	Use HashMap lookup with existing/missing-ID handling.	4/4	Requirement met in the submitted code and/or relevant evidence. No deduction.
4(e)	Duplicate prevention over maintained records.	Prevent duplicate participant IDs, workshop IDs, emails and participant/workshop pairs.	4/4	Requirement met in the submitted code and/or relevant evidence. No deduction.
4(f)	Wrapper conversions from entered text.	Use Integer and Double wrapper conversion meaningfully.	4/4	Requirement met in the submitted code and/or relevant evidence. No deduction.
4(g)	Fixed title sorting and array-based registration counting.	Implement requested sort/filter/count/income operations on stored records.	2/4	2/4: sorting the standard titles does not sort stored workshop records; demonstrate two requested operations meaningfully over the generic record collections.

Question	Student Answer Summary	Expected Answer / Marking Criteria	Marks Awarded	Feedback
4(h)	Reusable generic-collection processing method.	A reusable method accepts or processes a generic collection; a method-level <T> parameter is a valid example, not mandatory.	4/4	Requirement met in the submitted code and/or relevant evidence. No deduction.
Q4 total	Sum of assessed subquestions.	Question weighting: 32 marks.	**29/32**	4 + 3 + 4 + 4 + 4 + 4 + 2 + 4 = 29.
5(a)	Appropriate exception-handling structures.	Use relevant try/catch handling for application failures.	3/3	Requirement met in the submitted code and/or relevant evidence. No deduction.
5(b)	Main-menu and several capture errors are handled; some submenu/summary numbers are unguarded.	Cover menu/numeric/ID/registration/date/fee/capacity/quantity/file/database failures.	2/3	2/3: letters at the summary capacity prompt cause an uncaught InputMismatchException. Guard every numeric prompt, not only the main menu.
5(c)	Useful field, capacity and duplicate checks, but no booking-time past-date validation.	Enforce actual field, duplicate, date, active and full-workshop rules.	2/3	2/3: apply the workshop date and other booking rules consistently to local and SQL registrations.
5(d)	Helpful user-facing error messages.	Explain invalid input, missing records/files and connection/SQL failures clearly.	3/3	Requirement met in the submitted code and/or relevant evidence. No deduction.
5(e)	Recovery is provided in several paths but not all numeric operations.	Allow corrected input and continue safely after recoverable failures.	2/3	2/3: an isolated run terminated on abc at the summary capacity prompt. Catch the error, consume the invalid input and allow correction.
5(f)	Resources closed safely.	Use try-with-resources for files and appropriate JDBC resources.	3/3	Requirement met in the submitted code and/or relevant evidence. No deduction.
Q5 total	Sum of assessed subquestions.	Question weighting: 18 marks.	**15/18**	3 + 2 + 2 + 3 + 2 + 3 = 15.
6(a)	Reusable persistence and export methods.	Reusable registration writer, report exporter, error handler and workshop/participant loader.	3/3	Requirement met in the submitted code and/or relevant evidence. No deduction.
6(b)	Dynamic detailed report file.	Generate a report from actual registration records, not fixed demonstration rows.	3/3	Requirement met in the submitted code and/or relevant evidence. No deduction.
6(c)	Detailed report content and timestamp.	Include participant ID/name, workshop title/date, amount, status and generation timestamp.	3/3	Requirement met in the submitted code and/or relevant evidence. No deduction.
6(d)	Five summary fields are exported from stored arrays/counters.	Export workshop, participant and registration totals, available spaces and expected income.	2/3	2/3: initial workshop booking counts can differ from the actual stored registration pairs, so total registrations and available-space/income calculations need reconciliation.
6(e)	Buffered file reading returns/displays text lines.	Read/reconstruct workshop/participant records for continued application use.	1/3	1/3: it does not rebuild usable workshop/participant records, indexes and registration state. Parse the file into the application collections.
6(f)	Creation/export feedback is present; loading reports a text read.	Clearly report actual creation, reading, updates and export success/failure.	2/3	2/3: distinguish displaying file contents from restoring operational records. Report a usable load only once records have actually been reconstructed.

Question	Student Answer Summary	Expected Answer / Marking Criteria	Marks Awarded	Feedback
6(g)	Missing/unreadable files handled safely.	Show helpful feedback and keep the application running usefully.	3/3	Requirement met in the submitted code and/or relevant evidence. No deduction.
6(h)	Efficient buffered file techniques.	Use suitable buffered reading/writing and safe resource handling.	3/3	Requirement met in the submitted code and/or relevant evidence. No deduction.
Q6 total	Sum of assessed subquestions.	Question weighting: 24 marks.	**20/24**	3 + 3 + 3 + 2 + 1 + 2 + 3 + 3 = 20.
7(a)	Three-table relational schema and SQL reviewed.	Fields/types (6), primary/foreign keys (3), required constraints (3), SQL script (2): total 14.	14/14	14/14: fields/types 6/6, primary/foreign keys 3/3, required and unique constraints 3/3, creation script 2/2.
7(b)	Required JDBC operations and execution evidence.	Reusable connection; insert/read/search/update/cancel, joins/counts and successful connection/insert evidence.	4/4	Requirement met in the submitted code and/or relevant evidence. No deduction.
7(c)	Safe, documented JDBC implementation.	Prepared statements, safe resource closure, SQLException handling and documented connection settings.	4/4	Requirement met in the submitted code and/or relevant evidence. No deduction.
Q7 total	Sum of assessed subquestions.	Question weighting: 22 marks.	**22/22**	14 + 4 + 4 = 22.
8(a)	The major console actions are reachable.	All required actions work in the integrated application.	2/3	2/3: the load action displays text instead of restoring usable records, and summary entry is disconnected from the captured records.
8(b)	Some capture actions also insert SQL rows, but the stores diverge.	Input, collections, registrations, calculations, files and SQL use consistent records/counts.	1/3	1/3: clear resets counters without clearing maps/sets; loading does not rebuild records; SQL changes and local report counts are not consistently reconciled.
8(c)	Completed valid/invalid test evidence.	At least ten tests with input, expected/actual outcome, pass/fail and evidence reference.	3/3	Requirement met in the submitted code and/or relevant evidence. No deduction.
8(d)	Foreign-key and registration-confirmation fixes are described.	Two actual errors, causes, code corrections and successful retests.	2/3	2/3: moving registration insertion after confirmation is supported. For the foreign-key fix, a successful connection alone is not a retest that an orphan registration is rejected; include that before/after test.
8(e)	Reproducible README documentation.	Purpose, software/Java/DB versions, compilation/run/setup, file paths, limitations and assumptions.	3/3	Requirement met in the submitted code and/or relevant evidence. No deduction.
8(f)	Complete, logical submission package.	Assessment evidence, final source, SQL, README, samples/reports and testing documentation.	3/3	Requirement met in the submitted code and/or relevant evidence. No deduction.
Q8 total	Sum of assessed subquestions.	Question weighting: 18 marks.	**14/18**	2 + 1 + 3 + 2 + 3 + 3 = 14.

3 Detailed Feedback

Strengths
Good coverage of arrays, wrappers, strings, enum selection, collection storage, prepared JDBC operations and a completed ten-case Excel test table with evidence references.

Common mistakes
Parallel arrays and record snapshots become inconsistent, summaries ask for independent values, loading only displays text, and unguarded numeric prompts can terminate the program.

Topics needing improvement
Rebuild real records during loading; calculate summaries from those records; clear every related collection/index; recheck booking dates; and apply numeric recovery to all submenus.

Workings and verification
The two Java files compile on Java 17. The Excel test table and PDF evidence were reviewed alongside the source. A controlled summary input of abc reproduced an uncaught InputMismatchException. SQL structure and the safe connection/resource methods were inspected.

Notation and presentation
Typed code, screenshots, tables and console output were assessed. Use consistent identifiers, readable indentation, labelled dates and currency amounts. No unclear handwriting affected academic marking; no separate mathematical-notation task applies.

Mark calculation
Q1 11 + Q2 15 + Q3 15 + Q4 29 + Q5 15 + Q6 20 + Q7 22 + Q8 14 = 141/160. Percentage = (141/160) x 100 = 88.13%.

4 Final Lecturer Comment

A substantial attempt demonstrating many of the required fundamentals. Focus next on maintaining one consistent set of records and proving that loading, clearing, registration and reporting all use it safely.